High-Level Business Plan & Ecological Guide

Blue Ridge Stream Restoration & Mitigation Ventures

Co-Founder & Managing Director: Hunter (Fly Fishing Georgia North Mountains)
Board Member: Hadi Irvani (General Partner at Infill Capital Partners, University of Virginia - UVA)
Strategic Advisor & Sponsor: Hill Hardman (Granite Holdings)
Ultimate KPI: Fund world-class fly fishing trip to Argentina!

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1. Executive Summary

This business plan establishes the strategic launch of **Blue Ridge Stream Restoration & Mitigation LLC** (proposed name). The company represents a unique, highly profitable convergence of **advanced coldwater trout stream ecology** and **federal/state compensatory mitigation banking**.
By utilizing Natural Channel Design (NCD)—which employs native wood, logs, and root wads to restore degraded streams to their original geomorphic footings rather than allowing them to overwiden—Hunter will deliver permanent bank stabilization, massive biological diversity lift, and fast-tracked CWA 404 permit compliance for Southeast developers.

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2. Fluvial Geomorphology & Trout Life Ecology

To successfully market this business to private landowners and USACE regulators, Hunter must articulate the profound ecological science that drives our restoration design:

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                  COLDWATER TROUT ECOLOGY MATRIX

 [ WATER TEMP ] ■■■■■■■■■■■■■> [ SPAWNING BEDS ] ■■■■■■■■■■■■■> [ FOOD CHAIN ]
Ideal: < 65°F (18°C)       Clean, unsilted gravel         Abundant EPT insects
Lethal: > 68°F (20°C)      for egg oxygenation            (mayflies, caddisflies)
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The Essentials of Trout Survival

Trout are highly sensitive, indicator species. They require specific, high-quality habitat parameters to survive, spawn, and thrive:

1. **Coldwater Thermal Regime**: Trout are coldwater fish. Ideal water temperatures range from **50°F to 65°F**. Once water temperatures exceed **68°F**, trout experience severe thermal stress, dissolved oxygen levels plunge, and mortality rates spike.
2. **Dissolved Oxygen (DO)**: Trout require highly oxygenated water (minimum **7 mg/L**). Fast-flowing, tumbling water over instream rocky riffles is the natural aerator of the stream.
3. **Silt-Free Spawning Gravel**: Wild trout lay their eggs in gravel nests called "redds" located at the tails of pools and in shallow riffles. Water must flow freely through this gravel to supply oxygen to the developing embryos.
4. **Benthic Macroinvertebrates (The Trout Food Chain)**: Trout feed primarily on the aquatic larvae of three insect orders: **Ephemeroptera (Mayflies)**, **Plecoptera (Stoneflies)**, and **Trichoptera**

(Caddisflies)—collectively known as the EPT index. These insects only survive in clean, unsilted cobble and gravel substrates.

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3. The Threat: How Runoff Destroys Mountain Streams

Throughout North Georgia and the Southeast, historic logging, unpaved mountain roads (such as Peter Knob and Newport roads), and rapid commercial developments (retail centers and warehouses) are actively destroying trout streams through two primary mechanisms:

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                        THE ECOLOGICAL CASCADE

 [peak Runoff Flow] ■■■■> [Severe Bank Erosion] ■■■■> [Overwidened Channel]
                                    ■                         ■
                                    ▼                         ▼
                        [Siltation of Spawning]    [Solar Warming / Thermal]

                                    ■                         ■
                                    ■■■■■■■■■■■■■■■■■■■■■■■■■■■
                                                 ▼
                                   [Ecological Desert]
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1. Stormwater Runoff & Channel Overwidening

- **The Cause**: Deforestation and the creation of impervious surfaces (roofs, asphalt, compacted soils) prevent rainfall from absorbing into the ground. Instead, rainwater runs off instantly, creating extreme, high-volume peak flows during storms.
- **The Degradation**: These peak flows slam against fragile stream banks, tearing away soil and trees. Because the stream cannot access its natural floodplain to dissipate energy, the channel begins to erode laterally, **overwidening** from a stable $12\text{-foot} *bankfull channel to an unstable, shallow $25\text{-to-}30\text{-foot}* channel.
- **The Thermal Impact**: During summer, this wide, shallow sheet of water moves slowly and acts as a massive solar collector. Without tree canopy shade and deep pool structures, the water temperature easily climbs past the lethal 68°F threshold, turning a trout stream into an ecological desert.

2. Sediment Loading & Siltation

- **The Cause**: Erosion from construction sites, cleared buffers, and dirt roads washes massive amounts of sand, clay, and silt into the stream.
- **The Degradation**: This sediment settles in the slow-moving pools and riffles. It fills the interstitial spaces between cobbles and gravel, effectively **smothering trout spawning beds** and suffocating insect larvae. Sand and clay clog trout gills and obliterate the rocky substrate required for Mayflies and Caddisflies, completely wiping out the trout's food source.

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4. The Solution: Why Natural Channel Design (NCD) is the Best Restoration

Hunter's business utilizes **Natural Channel Design (NCD)** to reverse this degradation. Unlike traditional "hard civil engineering" (which uses concrete lining and rip-rap boulders that accelerate water flow, destroy ecosystems, and transfer erosion downstream), NCD uses **native wood, logs, root wads, and strategically placed boulders** to mimic natural geomorphic structures.
NCD represents the absolute **best type of stream restoration** because:

- **Rebuilds Stream Footings**: NCD physically narrows the bankfull channel back to its stable geomorphic cross-section (e.g., from an unstable $26\text{ ft} *width back to an optimal $12\text{ ft}* width). This increases water velocity in the center of the channel, naturally scouring out deep, cold pools.
- **Restores Wildlife & Biodiversity**: Re-establishing deep pool-riffle-run sequences provides instant habitat. Log structures and root wads introduce organic carbon into the stream, feeding the EPT insect population. Dissolved oxygen levels skyrocket as water tumbles over rock cross-vanes, creating a biological sanctuary.
- **Maximizes USACE SOP Credit Yield**: Because NCD provides massive "ecological lift" rather than simple cosmetic bank grading, it qualifies for the highest credit multipliers (**Priority 1 and Priority 2 Restoration**) under the USACE Savannah District SOP. This turns a single restored reach into a highly valuable credit asset.

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5. Corporate Entity & Governance Phased Roadmap

To manage risk and minimize launch costs, Hunter should execute a two-phase corporate structure:

- **Phase 1: Consulting Umbrella (Months 1–6)**: Establish a consulting division—**"Blue Ridge Stream Consulting"**—operating under Hunter's existing brand, *Fly Fishing Georgia North Mountains*. This leverages his established reputation, minimizes initial filing fees, and lets him use early showcase projects (Roya's Cabin and Hunter's Cabin) to build a visual track record.
- **Phase 2: Dedicated LLC Spin-Out (Months 6+)**: Form **Blue Ridge Stream Restoration & Mitigation LLC** as a separate legal entity. Heavy instream excavation requires heavy equipment and carries liability risk; a dedicated LLC isolates this risk. Furthermore, USACE MBIs and conservation easements require the bank sponsor to be a specific corporate entity with dedicated financial assurance accounts.

Board of Directors & Corporate Governance

The company will be guided by an active Board of Directors to ensure rigorous financial oversight, capital allocation, and regional commercial B2B developer scaling:

- **Hadi Irvani (Board Member)**: General Partner at Infill Capital Partners, Principal at Granite Holdings, University of Virginia (UVA). A leading Southeast real estate and logistics asset investor who provides corporate governance, capital structuring, and developer relationship management across major logistics corridors.
- **Hunter Morris (Co-Founder & Managing Director)**: Fluvial geomorphologist, Principal and Owner of *Fly Fishing North Georgia* (bookings@flyfishingnorthgeorgia.com). For over 15 years, Hunter has guided trophy fly-fishing excursions across pristine wild trout streams in Fannin and Gilmer counties. His deep daily observations of mountain stream siltation and lateral bank erosion led him to study advanced fluvial dynamics. As our geomorphic design-build lead, Hunter combines intimate hydraulic knowledge of streams with Natural Channel Design (NCD) and live soil bioengineering to deliver self-sustaining trout sanctuaries.

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6. Marketing, B2B Outreach, & Permitting Roadmap

Marketing & B2B Outreach

Hunter's outreach must be highly targeted, using the case studies as primary assets:

- **Private Landowners (The Showroom)**: Invite premium landowners and mountain land brokers (**Jack Chapman at Transwestern**, **Stream Realty Partners**) to **Hunter's Cabin at Goldmine Hollow** for a weekend of wild trout fishing on Noontootla Creek. Walk them through the log step-pools and sediment traps to demonstrate your bioengineering design.
- **Commercial Developers (The Credits)**: Target active regional developers (**Mike Irby at Taylor Mathis**, **Jeff Fuqua**, **Sefried**, **Ray Weeks**, **Maxwell Bonney**) who face stream piping or buffer variance hurdles. Propose supplying them with Priority 1 stream credits generated at **Roya's Cabin at Anderson Creek** (estimated **12,900 credits** total yield).
- **Mitigation Bankers**: Partner with **Neil Blackman (Principal at Corblu Ecology Group)** in a Joint Venture (JV). Corblu manages the banking ledger and RIBITS registry, while Hunter's company acts as the specialized design-build contractor.

The Regulatory Permitting Roadmap

Every project must navigate two parallel permitting tracks:

1. **Federal Pathway (USACE)**: Apply for a **Nationwide Permit 27 (NWP 27)** for Aquatic Habitat Restoration. This provides a fast-tracked, 60-to-90-day approval window for ecological restoration projects.
2. **State Pathway (Georgia EPD)**: Apply for a **State Stream Buffer Variance (SBV)** from EPD Environmental Engineer **Brian Kent** (`brian.kent@dnr.ga.gov`) for temporary stream buffer disturbance during construction.

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7. The Argentina Growth Target Dashboard

To ensure this business launches successfully and secures the ultimate prize (the Argentina fly fishing trip!), we must track these specific milestones:

Milestone	Target Objective	Timeline	Net Margin Contribution	Argentina Funding Tracker
Milestone 1	Permit Roya's Cabin (ACR) & Goldmine Hollow	Months 1–3	$0 (Upfront permitting)	■ **Stage 1**: Permitting filed
Milestone 2	Construct Anderson Creek & Goldmine Showcases	Months 4–6	$0 (Showcase asset creation)	■ **Stage 2**: Showcases completed
Milestone 3	Secure Neil Blackman (Corblu) JV & initial credit release	Months 7–12	**+$150,000+** (Initial 15% credit sale)	■ **Active Funding Secured!**

Milestone	Target Objective	Timeline	Net Margin Contribution	Argentina Funding Tracker
Milestone 4	Scale: Spin out LLC & close first corporate buyer (Fuqua/Sefried)	Months 12+	**+$250,000+** (Full credit liquidation)	■ **Business Fully Scaled**
Milestone 5	**Hadi & Hunter Fly Fishing in Patagonia, Argentina**	**Fall 2026**	**Fund $15,000 World-Class Trip**	■ **Goal: Giant Brown Trout!**